EXPERIMENT - 65

1. Problem Statement

Write a recursive program to solve the **Tower of Hanoi** problem for **n** disks and display all moves.

2. Algorithm

Main Program

Step 1: START
Step 2: Read the number of disks n.
Step 3: Call hanoi(n, 'A', 'C', 'B').
Step 4: STOP

Function hanoi(n, from, to, via)

Step 1: If n = 0, return.
Step 2: Call hanoi(n − 1, from, via, to).
Step 3: Display: Move disk n from from to to.
Step 4: Call hanoi(n − 1, via, to, from).

3. Test Case Table

#	n	Expected Moves (disk: from→to)	Actual Moves (disk: from→to)	Result
1	1	1: A→C	1: A→C	**PASS**
2	2	1: A→B, 2: A→C, 1: B→C	1: A→B, 2: A→C, 1: B→C	**PASS**
3	3	1: A→C, 2: A→B, 1: C→B, 3: A→C, 1: B→A, 2: B→C, 1: A→C	1: A→C, 2: A→B, 1: C→B, 3: A→C, 1: B→A, 2: B→C, 1: A→C	**PASS**

4. Source Code

hanoi65.c

```
/* Experiment 65: Tower of Hanoi using recursion. */
#include <stdio.h>

void hanoi(int n, char from, char to, char via);

int main(void)
{
    int n;

    printf("Enter the number of disks: ");
    scanf("%d", &n);

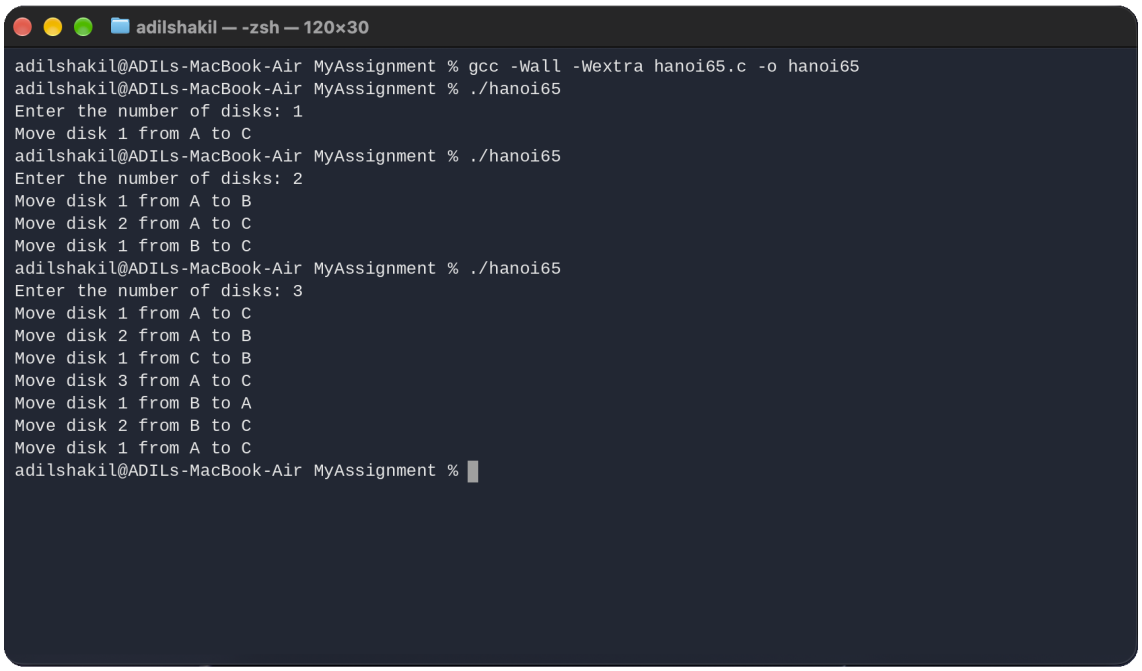
    hanoi(n, 'A', 'C', 'B');
    return 0;
}

/* Move n-1 disks out of the way onto via, move the biggest disk to
   its place, then bring the n-1 disks back on top of it. */
void hanoi(int n, char from, char to, char via)
{
    if (n == 0)
        return;

    hanoi(n - 1, from, via, to);
    printf("Move disk %d from %c to %c\n", n, from, to);
    hanoi(n - 1, via, to, from);
}
```

5. Compilation and Output

Compiled with `gcc -Wall -Wextra hanoi65.c -o hanoi65` so the small warnings would show up too. It built clean.



All three passed.